

Final Dissertation Project

Introduction, CRISP-DM Framework, and Topic Selection

BSc (Honours) Data Science | Year 3 Briefing

Module Coordinators

Dr. Zaid Al-Huda · Dr. Pedro Mestre

Stirling College, Chengdu University

Today's Briefing

What we will cover today

01

About the Module

Overview, credits,
key rules

02

CRISP-DM Framework

Six-phase lifecycle for
your project

03

Standards & Integrity

Critical thinking and
AI policy

04

Timeline & Next Steps

Full year plan to your
final viva

05

Project Selection

Choose, rank, and get
started

PART 01

About the Module

CSCU9Z7 is the capstone of your BSc Data Science degree.

What is CSCU9Z7?

The Final Dissertation module for BSc (Honours) Data Science - 60 credits spanning the entire final year.

60 CREDITS
Full Final Year

AT A GLANCE

Dissertation Outline

Nov 2026 (TBC)

10 % of total mark · 3,000 words

Final Thesis

April 2027 (TBC)

90 % of total mark · 10,000 words

Final Defence (Viva)

April 2027 (TBC)

Compulsory · live code demo

Similarity Limit

2–10 %

Turnitin threshold

Ethics Approval

Autumn Wk 4, 2026

Assess ethical implications and obtain committee approval.

Project Proposal

Autumn Wk 3, 2026

Agree your topic with your supervisor before the deadline.

Why It Matters

Your dissertation is more than a final exam.



Proof of Skill

Employers and graduate schools read dissertations. It proves you can scope, build, and defend a complete data science project end to end.



Independent Work

You move from following lab tasks to leading a project. You make all design decisions and justify every one with evidence.



Critical Thinking

Higher marks come from analysis, not description. Compare approaches, weigh trade-offs, and explain WHY at every step.



Research Foundation

A strong dissertation often becomes a portfolio piece or a stepping stone into a master's programme or career opportunity.

Your Commitment

Weekly supervisor meetings are not optional.

Missing two or more meetings without a valid reason will result in the project failing.

20 Meetings Total

12 meetings in Semester 1, 8 in Semester 2.
One individual meeting with your supervisor every single week.

Attendance Policy

Notify your supervisor in advance if you cannot attend. No reason given counts as unexcused.
Two unexcused = fail.

What to Bring

Questions, evidence of progress, and draft material. Arriving empty-handed is not acceptable at this level.

PART 02

The CRISP-DM Framework

Cross-Industry Standard Process for Data Mining - the lifecycle guiding your project.

Why Follow a Framework at All?

Three reasons CRISP-DM will make your dissertation stronger.

01

It Prevents Skipping Steps

Without a framework, students rush to modelling and produce weak results because the data was never properly understood.

02

It Maps to Your Dissertation

Each phase becomes a chapter or section. CRISP-DM gives you a ready-made writing structure for the whole thesis.

03

It Is Iterative by Design

You will return to earlier phases as you learn. The framework expects this and guides you on how to do it properly.

MOST USED

CRISP-DM

is the most widely-used process model for data science in industry.

Created 1996. Still the global standard.
Used by IBM, Microsoft, healthcare, banking, and research labs worldwide.

CSCU9Z7 Final Dissertation

CRISP-DM Project Blueprint · Every phase, every task, one clear purpose

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01 Business Understanding

- Task 1.1** Define research question & objectives
- Task 1.2** Literature review & gap analysis
- Task 1.3** Write project proposal & plan
- Task 1.4** Identify stakeholders & success criteria

02 Data Understanding

- Task 2.1** Identify & collect relevant datasets
- Task 2.2** Perform exploratory data analysis (EDA)
- Task 2.3** Assess data quality & completeness
- Task 2.4** Document data dictionary & sources

03 Data Preparation

- Task 3.1** Clean, transform & normalise data
- Task 3.2** Handle missing values & outliers
- Task 3.3** Feature engineering & selection

YOUR
PROJECT

Iterative cycle - revisit earlier phases as your dissertation evolves

04 Modelling

- Task 4.1** Select appropriate ML/DL algorithms
- Task 4.2** Train models & tune hyperparameters
- Task 4.3** Apply cross-validation techniques
- Task 4.4** Compare baseline vs. proposed models

05 Evaluation

- Task 5.1** Evaluate with metrics (accuracy, F1, etc.)
- Task 5.2** Interpret results vs. research objectives
- Task 5.3** Assess ethical & practical implications

06 Deployment

- Task 6.1** Write final dissertation report
- Task 6.2** Build demo / prototype (if applicable)
- Task 6.3** Prepare viva presentation & defence
- Task 6.4** Publish results & suggest future work

Dissertation Structure

Every chapter, every mark, one clear purpose

01 Abstract & Introduction

- Specific problem statement supported by evidence
- SMART objectives with clearly defined scope

CRISP-DM Phase 01 - Business Understanding

02 Literature Review

- Thematic grouping - NOT paper-by-paper listing
- Gaps clearly identified and linked to this project

CRISP-DM Phase 01 - Business Understanding

03 Data Analysis

- Dataset description, EDA and quality assessment
- Preprocessing steps and feature engineering

CRISP-DM Phase 02 + 03 - Data Understanding & Preparation

04 Methodology

- Architecture, tools and technical choices justified
- Alternatives explored - WHY this approach?

CRISP-DM Phase 04 - Modelling

05 Evaluation

- Metrics justified - benchmarked against baselines
- Critical interpretation - WHERE and WHY it works

CRISP-DM Phase 05 - Evaluation

06 Conclusion & Future Work

- Summary of contributions and key findings
- Limitations acknowledged + future directions

CRISP-DM Phase 06 - Deployment

PHASE 1

Business Understanding

Why are we doing this project at all? Define the problem before you touch the data.

Maps to: Chapter 1 Introduction & Abstract

ACTIVITIES

- State the real-world problem in clear language
- Identify who suffers if it remains unsolved
- Translate the problem into a data science task
- Write SMART research objectives
- Define success criteria with specific numbers

COMMON MISTAKES

- Generic statements like 'diabetes is serious'
- No measurable success criteria stated upfront
- Confusing a tool (e.g. CNN) with the problem
- Skipping straight to picking a model
- No statistic or reference to justify the problem

Phase 2: Data Understanding

Get to know your data deeply before you transform it. Curiosity now saves time later.

ACTIVITIES

- Identify and cite the dataset source clearly
- Describe features, target variable, and data types
- Compute descriptive statistics (mean, median, SD)
- Plot distributions and check for outliers
- Investigate missing values and class imbalance

CRITICAL QUESTIONS TO ASK

- Is the data large enough to train your model?
- Are classes balanced or severely skewed?
- Are there obvious correlations or leakage risks?
- How was the data collected - is there bias?
- Are there ethical or licensing concerns?

Phase 3: Data Preparation

Often 60–80 % of real project time. Quality of preparation limits quality of results.

ACTIVITIES

- Clean data: remove or impute missing values
- Encode categorical features sensibly
- Scale or normalise numeric features where needed
- Split into train, validation, and test sets
- Handle imbalance: SMOTE, class weights
- Engineer new features when justified by domain

JUSTIFY EACH CHOICE

- Why mean imputation, not median or model-based?
- Why one-hot encoding, not target encoding?
- Why this train/test split ratio (e.g. 70/15/15)?
- Why oversample rather than undersample?
- Every choice needs a reason AND a reference
- Justify in writing as you work - memory fades

PHASE 4

Modelling

*Build models, tune them, and document everything.
Bring more than one model to the comparison.*

Maps to: Chapter 4 Methodology

ACTIVITIES

- Select candidate models with clear reasoning
- Train a baseline first, then improve step by step
- Tune hyperparameters systematically (grid/random)
- Use cross-validation, not single train/test splits
- Track experiments, seeds, and configs for reproducibility

ALWAYS COMPARE ALTERNATIVES

- Train at least two or three different models
- Include a simple baseline (e.g. logistic regression)
- Discuss what each model is good and bad at
- Justify the final model with evidence, not preference
- Document alternatives you rejected and why

Phase 5: Evaluation

Did you actually solve the problem? Evaluation is where the marker checks your honesty.

Activities

- Use the metrics you justified upfront
- Build a comparison table of all models
- Inspect confusion matrix and error patterns
- Compare against at least three published papers
- Discuss WHERE the model fails and WHY

What Markers Look For

- Metric choice is justified, not assumed
- Honest discussion of weaknesses and limits
- Specific numerical comparison with literature
- Statistical significance where relevant
- Clear connection back to Chapter 1 objectives

Phase 6: Deployment / Reporting

In industry: deploy the model. In your dissertation: clean code, a README, and a strong final report.

ACTIVITIES

- Package code with a clear README file
- Document setup, run, and reproduction steps
- Write dissertation chapters in academic style
- Prepare for the viva: know every choice made
- Reflect honestly on what you would do differently

DEPLOYMENT IN YOUR DISSERTATION

- Working demo ready for the final code viva
- ZIP archive that markers can run immediately
- Visualisations and results clearly presented
- Limitations and future work discussed honestly
- Critical reflection on the full process itself

CRISP-DM Is a Loop, Not a March

Real projects revisit earlier phases. Plan for it. A dissertation that shows revision scores higher than one that hides it.

Loop 1

Modelling Reveals Data Issues

Your model is overfitting. You return to Phase 3 for more cleaning or Phase 2 to investigate feature leakage.

Loop 2

Evaluation Reveals Weak Metrics

Accuracy is high but recall is poor. You return to Phase 1 to confirm the metric was the right choice for the problem.

Loop 3

EDA Reveals New Questions

Surprising patterns emerge. You return to Phase 1 to refine your research question to match what the data can answer.

Treat iteration as a strength - not a sign of failure

PART 03

Standards & Integrity

Two non-negotiable standards: critical thinking and academic honesty.

Critical Thinking: Description Loses Marks

80 %

of students only describe

40–55 %

marks for description alone

60 %+

marks for real analysis

DESCRIPTION - LOW MARKS

"Paper X uses CNN and got 88% accuracy."

No comparison, no context, no WHY. The marker cannot tell whether you understood it.

ANALYSIS - HIGHER MARKS

"Paper X: 88 % with CNN but needs 50 k images. Paper Y: 85 % via transfer learning on 2 k. My project targets small-data regimes - bridging this gap."

Test every paragraph: did I just say WHAT - or did I also explain WHY and SO WHAT?

Academic Honesty & AI Policy

Plagiarism and AI similarity scores must remain between 2 and 10 %. Higher scores trigger investigation.

AI MAY HELP WITH

- Brainstorming chapter structure ideas
- Suggesting outlines for sections
- Checking grammar and clarity of your writing
- Explaining unfamiliar concepts to you
- Polishing sentences you have already written

AI MUST NOT WRITE

- Your dissertation content or analysis
- Your literature review or critical review
- Your code or data interpretation
- Your conclusions or critical reflection
- Anything submitted without your own editing

Policy level: AIAS Level 2. All AI use must be declared in your code README.

PART 04

Timeline and Next Steps

Working backwards from the deadline gives you confidence, not panic.

The Full Year

From topic selection to your final viva

May – Jun

Topic Selection

Topic list released. Browse the list and contact supervisors up to Week 13. .

Jul – Aug

Summer Prep

Read review papers in your field. Explore datasets. Come September prepared.

Sep – Oct

Semester 1: Planning

First meeting with supervisor. Project proposal due Week 3, Oct 2026 (TBC). Ethics approval due Week 4.

Nov 2026

Dissertation Outline

3,000-word outline due Nov 2026 (exact date TBC). This counts as 10% of your total mark.

Nov – Dec

Technical Work

Data analysis, modelling, and evaluation. All technical work should be complete by end of December.

Jan – Feb

Writing

Use your results to write all chapters. Supervisor feedback on up to 3,000 words of draft.

Mar 2027

Pre-Defense (Code Demo)

Interim demo session (Spring Week 1, 2027). Show progress and get peer feedback.

Apr 2027

Final Submission

Dissertation PDF submitted. Counts as 90% of your total mark. Exact date TBC.

Apr 2027

Final Viva

Live code demo with supervisor and second marker. Failing the viva means failing the project. Exact date TBC.

Official Deadlines

Every milestone from the module handbook

Every Week	Weekly Supervisory Meetings Talk to your supervisor, update on progress, ask questions. Very strongly recommended.	0%
Autumn Wk 3, 2026	Project Proposal Agree your project with your supervisor before the Autumn semester final deadline.	0%
Autumn Wk 4, 2026	Ethics Approval Assess ethical implications of your project and obtain ethics approval from the committee.	0%
Nov 2026 (TBC)	Dissertation Outline 3,000 words. Technical content 70%, Presentation 30%.	10%
Spring Wk 1, 2027	Pre-Defense (Code Demo) Formative only. Practice your live demo and get feed-forward from the panel.	0%
Spring Wk 3, 2027	Poster & Pitch Session Formative only. Communicate your project to peers, staff, and employers.	0%
Spring Wk 6, 2027	Draft to Supervisor Optional but strongly recommended. Submit at least 2 weeks before the final deadline.	0%
April 2027 (TBC)	Final Dissertation 10,000 words. Technical 80%, Presentation 20%. Submit PDF and code ZIP separately.	90%
April 2027 (TBC)	Final Defense (Viva) Compulsory. Live code demo with supervisor and second marker. Fail viva = fail module.	0%*

Tips That Work

What separates strong dissertations from weak ones

01 Justify everything

For every choice, write the WHY in one sentence before you forget it.

02 Compare, do not list

In your literature review, group papers by theme - not by paper.

03 Track your experiments

Save every run, every seed, every config. Reproducibility wins marks.

04 Write while you build

Draft Chapter 3 as you do EDA. Memory fades faster than you think.

05 Read three papers fully

Compare your final results with at least three published works in your area.

06 Meet your supervisor weekly

Bring questions, drafts, and progress. Two missed meetings = a fail.

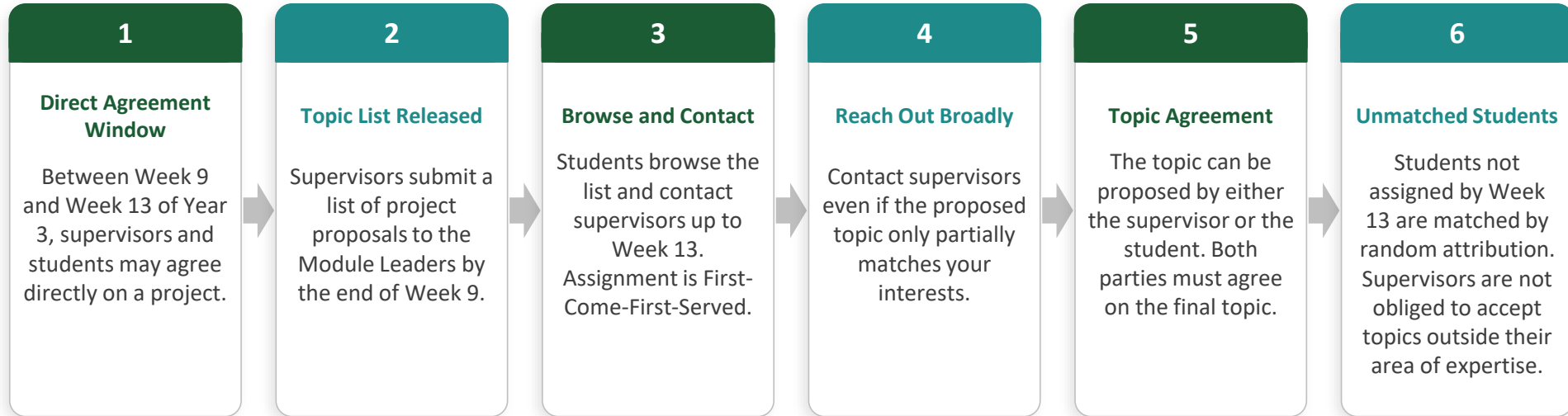
PART 05

Project Selection

Choose a problem you can defend, not just a topic that sounds impressive.

How It Works

The selection process at a glance



Choosing Wisely

Five questions to ask before you rank a topic

Q

Can I clearly state the problem in one sentence?

If you cannot, the topic is too vague or you do not understand it yet. Read more before ranking.

Q

Is there real data I can access right now?

Look for public datasets on Kaggle, UCI, or government portals. No data, no project.

Q

Can I find at least three published papers on this?

You will need to compare your results to literature. If the area is too niche, you will struggle.

Q

Do I have the technical skills, or can I learn them?

Be honest. A familiar method done well beats a fashionable method done badly.

Q

Will I still want to work on this in six months?

Motivation matters. Pick a topic you can talk about with energy in your viva.

Questions?

Use the next 5 minutes - this is the cheapest help you will get all year.

YOUR NEXT STEPS

- Browse the supervisor topic list released at the end of Week 9 of Year 3.
- Contact supervisors directly between Week 9 and Week 13 to agree on a project.
- Reach out even if the topic only partially matches your interests - it is First-Come-First-Served.
- Agree the final topic with your supervisor - both parties must confirm before the Week 13 deadline.
- If unmatched by Week 13, you will be assigned a supervisor by random attribution.